

R Programming

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Heart Disease Patient Data Analysis

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1. Introduction

Heart disease is one of the leading causes of death worldwide, and identifying patterns in patient health data can help in early detection and prevention.

This project analyzes the Heart Disease dataset (originally from the UCI Machine Learning Repository and widely used on Kaggle as the “Heart Disease UCI” dataset) using R programming. The analysis involves importing the dataset, cleaning and pre-processing the data, performing statistical analysis, and creating meaningful graphs and visualizations.

The main purpose of this project is to identify important clinical patterns, understand chest pain type and diagnostic test performance, analyze the presence of heart disease, and discover useful relationships between patient variables. The results can help illustrate how statistical analysis and visualization support data-driven health insights.

2. Objectives

The main objectives of this project are:

- To import and understand the heart disease dataset using R.
- To clean and preprocess the dataset before analysis.
- To check for missing values and data inconsistencies.
- To identify the most common chest pain types among patients.
- To analyze the distribution of thalassemia test results.
- To study cholesterol level distribution across patients.
- To examine the relationship between age and maximum heart rate.
- To create meaningful graphs and visualizations using R.
- To interpret the results and identify important clinical insights.

3. Data Importing

The Heart Disease dataset used in this project contains information about patient age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting ECG results, maximum heart rate achieved, exercise-induced angina, ST depression, slope, number of major vessels, thalassemia test result, and heart disease diagnosis.

The dataset is imported into R using the `read.csv()` function. After importing the data, the first few

observations are displayed to verify that the dataset has been loaded correctly.

```
heart_data<-read.csv("heart_disease_dataset.csv")
head(heart_data)
```

```
##  age sex cp trestbps chol fbs restecg
## 1  63  1  1      145  233  1      2
## 2  67  1  4      160  286  0      2
## 3  67  1  4      120  229  0      2
## 4  37  1  3      130  250  0      0
## 5  41  0  2      130  204  0      2
## 6  56  1  2      120  236  0      0

##  thalach exang oldpeak slope ca      thal target
## 1      150    0      2.3    3  0    fixed      0
## 2      108    1      1.5    2  3    normal     1
## 3      129    1      2.6    2  2 reversible  0
## 4      187    0      3.5    3  0    normal     0
## 5      172    0      1.4    1  0    normal     0
## 6      178    0      0.8    1  0    normal     0
```

The dimensions of the dataset are checked to determine the total number of observations and variables.

```
dim(heart_data)
```

```
## [1] 247 14
```

The names of all variables in the dataset are displayed to understand the information available for analysis.

```
names(heart_data)
```

```
## "age" "sex" "cp" "trestbps" "chol" "fbs" "restecg"
## "thalach" "exang" "oldpeak" "slope" "ca" "thal" "target"
```

4. Understanding the Dataset

```
str(heart_data)
```

```
## 'data.frame':      247 obs. of  14 variables:
##  $ age      : int  63 67 67 37 41 56 62 57 63 53 ...
##  $ sex      : int  1 1 1 1 0 1 0 0 1 1 ...
##  $ cp       : int  1 4 4 3 2 2 4 4 4 4 ...
##  $ trestbps: int  145 160 120 130 130 120 140 120 130 140 ...
##  $ chol     : int  233 286 229 250 204 236 268 354 254 203 ...
##  $ fbs      : int  1 0 0 0 0 0 0 0 0 1 ...
##  $ restecg  : int  2 2 2 0 2 0 2 0 2 2 ...
##  $ thalach  : int  150 108 129 187 172 178 160 163 147 155 ...
##  $ exang    : int  0 1 1 0 0 0 0 1 0 1 ...
##  $ oldpeak  : num  2.3 1.5 2.6 3.5 1.4 0.8 3.6 0.6 1.4 3.1 ...
```

```
## $ slope : int 3 2 2 3 1 1 3 1 2 3 ...
## $ ca : int 0 3 2 0 0 0 2 0 1 0 ...
## $ thal : chr "fixed" "normal" "reversible" "normal" ...
## $ target : int 0 1 0 0 0 0 1 0 1 0 ...
```

```
summary(heart_data)
```

```
##      age      sex      cp      trestbps
## Min.   :29.00 Min.   :0.00 Min.   :1.00 Min.   :94.00
## 1st Qu.:47.50 1st Qu.:0.00 1st Qu.:2.00 1st Qu.:120.00
## Median :56.00 Median :1.00 Median :3.00 Median :130.00
## Mean   :54.70 Mean   :0.66 Mean   :3.14 Mean   :132.00
## 3rd Qu.:62.00 3rd Qu.:1.00 3rd Qu.:4.00 3rd Qu.:140.00
## Max.   :77.00 Max.   :1.00 Max.   :4.00 Max.   :200.00

##      chol      fbs      restecg      thalach
## Min.   :126.00 Min.   :0.00 Min.   :0.00 Min.   :71.00
## 1st Qu.:212.00 1st Qu.:0.00 1st Qu.:0.00 1st Qu.:132.00
## Median :243.00 Median :0.00 Median :0.00 Median :153.00
## Mean   :247.39 Mean   :0.14 Mean   :0.94 Mean   :149.19
## 3rd Qu.:275.50 3rd Qu.:0.00 3rd Qu.:2.00 3rd Qu.:167.50
## Max.   :564.00 Max.   :1.00 Max.   :2.00 Max.   :202.00

##      exang      oldpeak      slope      ca
## Min.   :0.00 Min.   :0.00 Min.   :1.00 Min.   :0.00
## 1st Qu.:0.00 1st Qu.:0.00 1st Qu.:1.00 1st Qu.:0.00
## Median :0.00 Median :0.70 Median :2.00 Median :0.00
## Mean   :0.33 Mean   :1.02 Mean   :1.60 Mean   :0.67
## 3rd Qu.:1.00 3rd Qu.:1.60 3rd Qu.:2.00 3rd Qu.:1.00
## Max.   :1.00 Max.   :6.20 Max.   :3.00 Max.   :3.00

##      target
## Min.   :0.00
## 1st Qu.:0.00
## Median :0.00
## Mean   :0.27
## 3rd Qu.:1.00
## Max.   :1.00
```

5. Checking Missing Values

Data cleaning is an essential step to ensure there are no missing values, incorrect data formats, or inconsistencies before running the analysis.

```
colSums(is.na(heart_data))
```

```
## age sex cp trestbps chol fbs restecg
## 0 0 0 0 0 0 0
## thalach exang oldpeak slope ca thal target
## 0 0 0 0 0 0 0
```

6. Data Cleaning

The dataset uses numeric codes for some categorical variables. These are converted into readable labels using `factor()` to make the analysis and charts easier to interpret.

```
heart_data$cp_label <- factor(
  heart_data$cp,
  levels = c(1, 2, 3, 4),
  labels = c("Typical Angina", "Atypical Angina",
             "Non-Anginal Pain", "Asymptomatic")
)

heart_data$sex_label <- factor(
  heart_data$sex,
  levels = c(0, 1),
  labels = c("Female", "Male")
)

heart_data$target_label <- factor(
  heart_data$target,
  levels = c(0, 1),
  labels = c("No Disease", "Disease Present")
)

head(heart_data[, c("cp_label", "sex_label", "target_label")])
```

```
##      cp_label sex_label target_label
## 1 Typical Angina      Male    No Disease
## 2 Asymptomatic      Male Disease Present
## 3 Asymptomatic      Male    No Disease
## 4 Non-Anginal Pain      Male    No Disease
## 5 Atypical Angina    Female    No Disease
## 6 Atypical Angina      Male    No Disease
```

The dataset contains no missing values, so no rows needed to be removed during cleaning.

7. Cholesterol Analysis

Cholesterol statistics are calculated using the cholesterol level (mg/dl) reported for each patient.

```
mean(heart_data$chol, na.rm = TRUE)
```

```
## [1] 247.3927
```

```
median(heart_data$chol, na.rm = TRUE)
```

```
## [1] 243.0
```

```
var(heart_data$chol, na.rm = TRUE)
```

```
## [1] 2593.7923
```

```
sd(heart_data$chol, na.rm = TRUE)
```

```
## [1] 50.9293
```

Interpretation

The mean represents the average cholesterol level across all patients in the dataset, while the median represents the middle cholesterol value. The difference between the mean and median indicates that the cholesterol distribution is affected by a few unusually high observations.

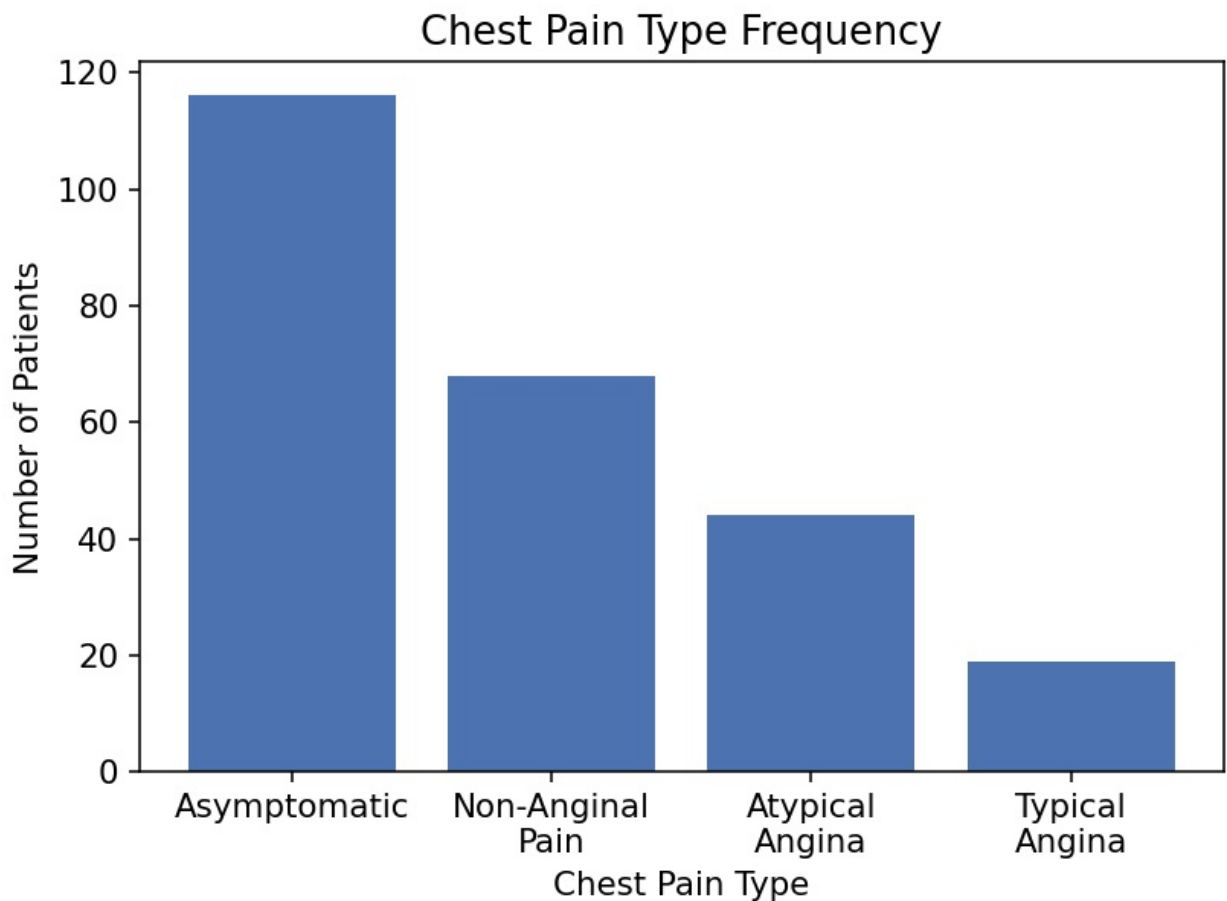
8. Chest Pain Type Analysis

```
cp_count <- table(heart_data$cp_label)
sort(cp_count, decreasing = TRUE)
```

```
## Asymptomatic Non-Anginal Pain Atypical Angina Typical Angina
##           116           68           44           19
```

9. Chest Pain Type Visualization

```
barplot(
  sort(cp_count, decreasing = TRUE),
  main = "Chest Pain Type Frequency",
  xlab = "Chest Pain Type",
  ylab = "Number of Patients",
  las = 2
)
```



Interpretation

The bar chart shows the number of patients reporting each chest pain type. Asymptomatic chest pain is the most frequently recorded type in the dataset (116 patients), followed by non-anginal pain (68), atypical angina (44) and typical angina (19). This helps identify which chest pain presentations are most common among the patients in this dataset.

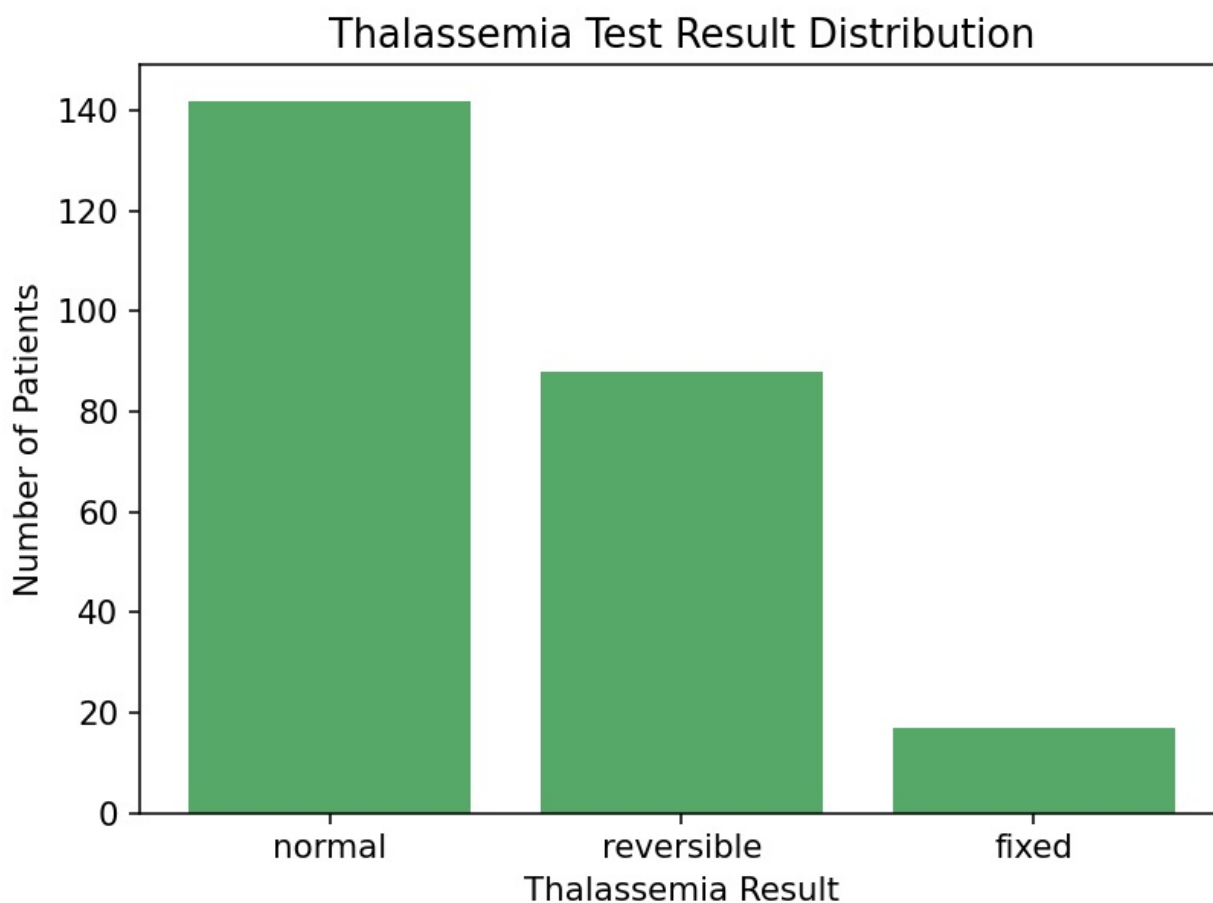
10. Thalassemia Test Result Analysis

```
thal_count <- table(heart_data$thal)
thal_count
```

```
## fixed normal reversible
##    17    142     88
```

```
barplot(
  thal_count,
  main = "Thalassemia Test Result Distribution",
  xlab = "Thalassemia Result",
  ylab = "Number of Patients",
  las = 2
```

)



Interpretation

The bar chart shows the distribution of thalassemia test results across the patients in the dataset. A “normal” result is the most common outcome (142 patients), followed by “reversible” (88) and “fixed” (17) defects. This shows that most patients in the dataset returned a normal thalassemia test result.

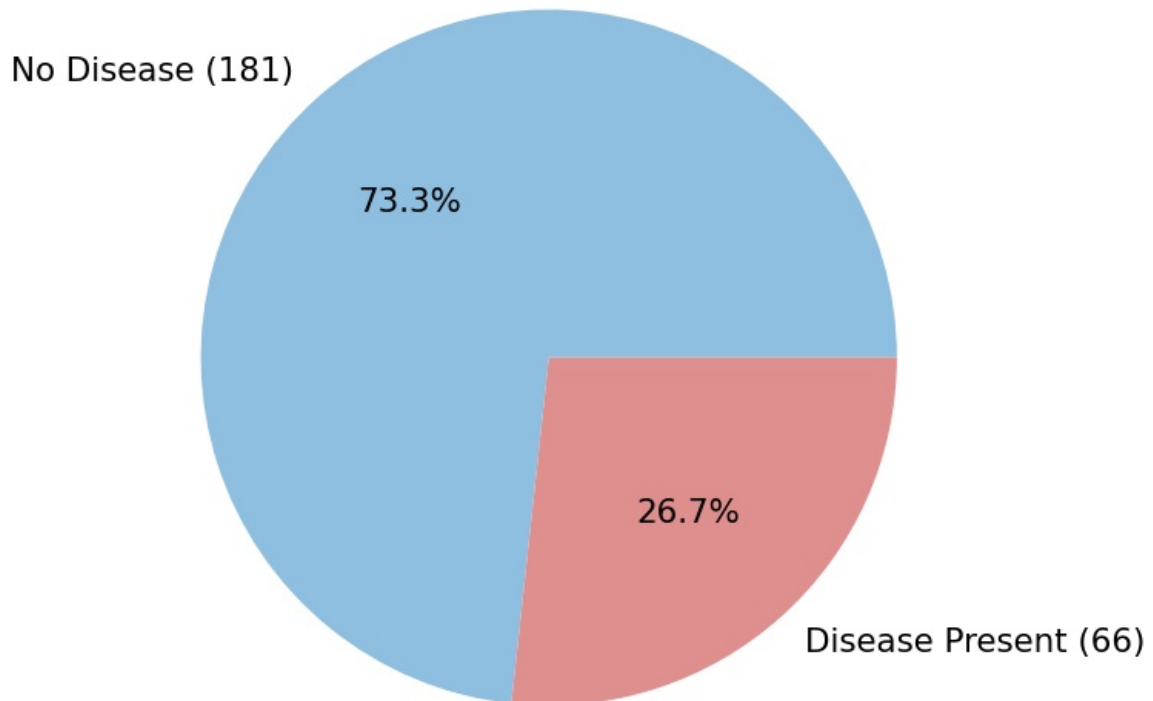
11. Heart Disease Diagnosis Analysis

```
target_count <- table(heart_data$target_label)
target_count
```

```
## Disease Present No Disease
##           66          181
```

```
pie(
  target_count,
  main = "Heart Disease Presence vs Absence",
  labels = paste(names(target_count), target_count)
)
```

Heart Disease Presence vs Absence

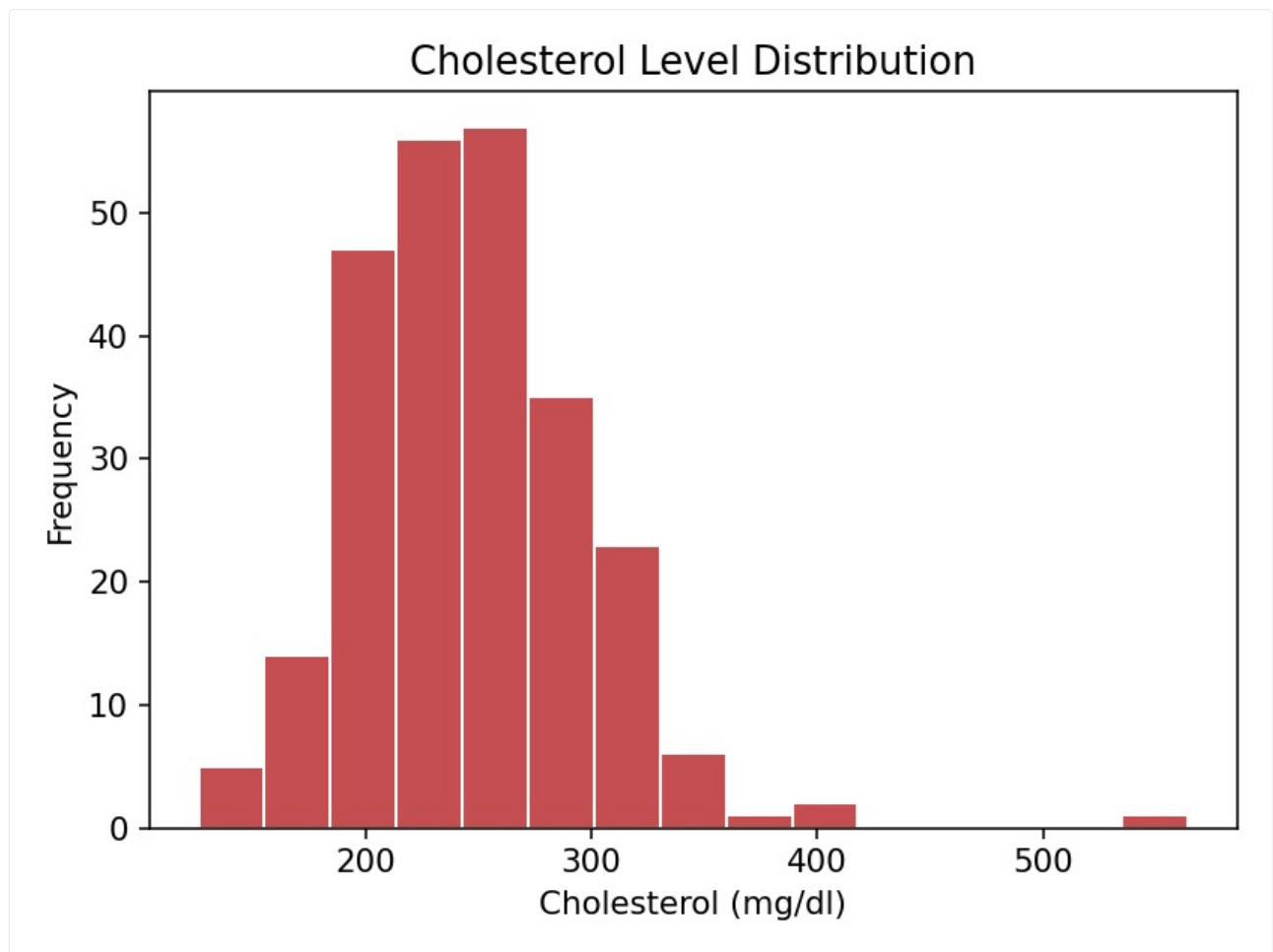


Interpretation

The pie chart compares patients diagnosed with heart disease against those without it. Out of 247 total patient records, 66 are identified as having heart disease, while 181 are classified as not having the condition. Therefore, patients without heart disease form the larger portion of this dataset.

12. Cholesterol Distribution

```
hist(  
  heart_data$chol,  
  main = "Cholesterol Level Distribution",  
  xlab = "Cholesterol (mg/dl)",  
  ylab = "Frequency"  
)
```

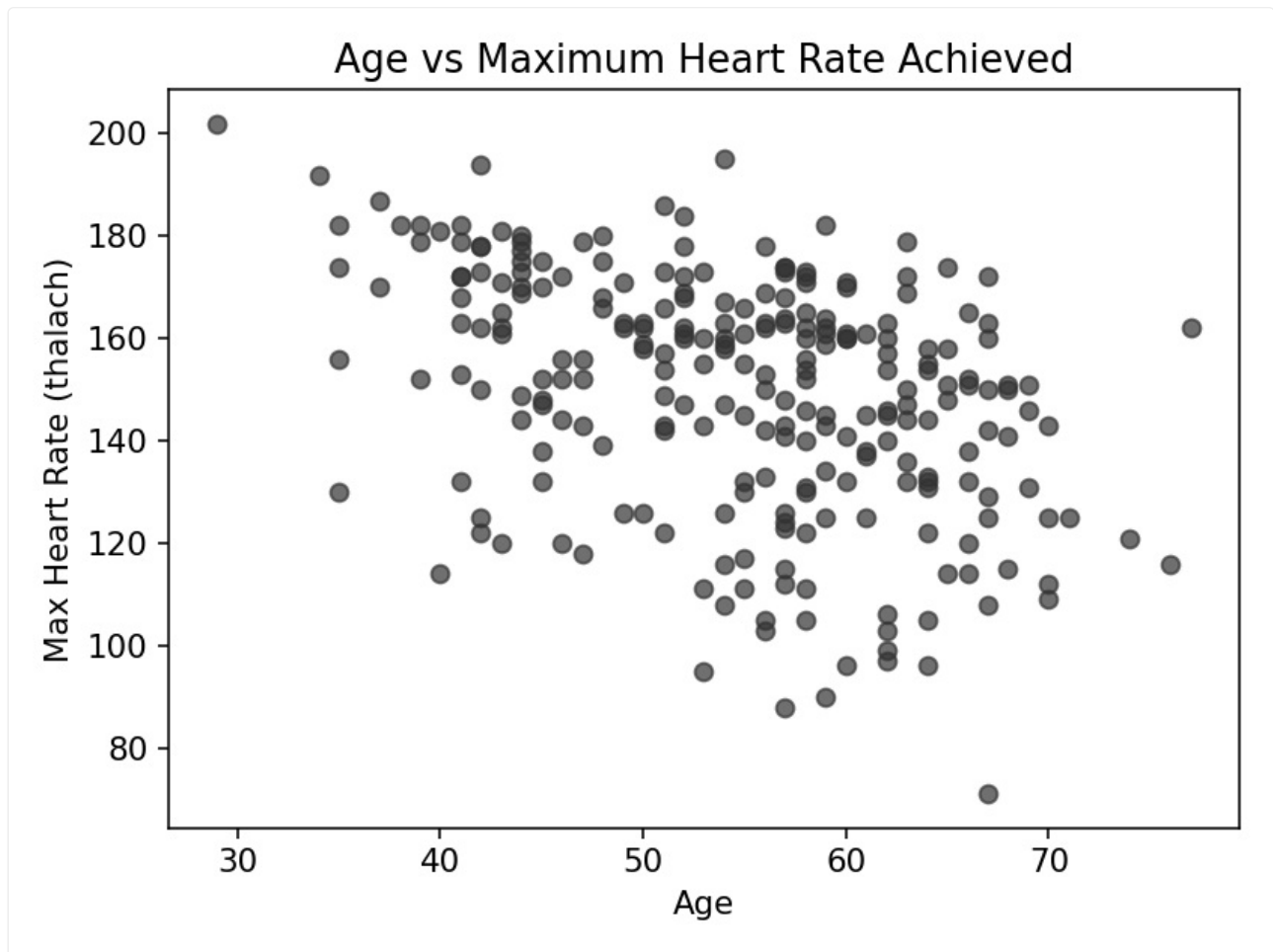


Interpretation

The histogram shows the distribution of cholesterol levels reported for the patients. Most observations are concentrated in a moderate cholesterol range, while a small number of patients have considerably higher cholesterol values. This indicates that the cholesterol distribution is right-skewed and contains some high-value observations.

13. Age vs Maximum Heart Rate Analysis

```
plot(
  heart_data$age,
  heart_data$thalach,
  main = "Age vs Maximum Heart Rate Achieved",
  xlab = "Age",
  ylab = "Maximum Heart Rate (thalach)",
  pch = 19
)
```



Interpretation

The scatter plot shows the relationship between a patient's age and the maximum heart rate achieved during testing. The points show a general downward pattern, indicating that older patients tend to achieve a lower maximum heart rate than younger patients.

14. Correlation Analysis

```
cor(  
  heart_data$age,  
  heart_data$thalach,  
  use = "complete.obs"  
)
```

```
## [1] -0.4098019
```

Interpretation

The correlation coefficient between age and maximum heart rate is approximately -0.4098019. This indicates a moderate negative relationship between the two variables. This means that as a patient's age increases, their maximum heart rate tends to decrease. However, correlation does not imply causation.

15. Findings

Based on the analysis of the Heart Disease dataset, the following findings were obtained:

- 1) The dataset contains 247 patient records with no missing values across any of the 14 variables.
- 2) Asymptomatic chest pain is the most frequently reported chest pain type among patients, followed by non-anginal pain, atypical angina and typical angina.
- 3) A "normal" thalassemia test result is the most common outcome among patients, followed by "reversible" and "fixed" defects.
- 4) 66 out of 247 patients are diagnosed with heart disease, while 181 are not.
- 5) The cholesterol distribution shows that most patients have moderate cholesterol levels (mean 247.3927 mg/dl, median 243.0 mg/dl), with a small number of unusually high observations.
- 6) The correlation coefficient between age and maximum heart rate is approximately -0.4098019, indicating a moderate negative relationship between the two variables.
- 7) The visualizations provide useful insights into chest pain types, thalassemia results and cholesterol patterns among heart disease patients.

16. Conclusion

This project analyzed a real-world Heart Disease dataset using R programming.

The analysis included data importing, data exploration, missing-value analysis, data cleaning, statistical analysis and data visualization.

The study examined chest pain types, thalassemia test results, cholesterol levels and the relationship between age and maximum heart rate among patients. It also compared the presence and absence of heart disease across the dataset.

The correlation analysis showed a moderate negative relationship between age and maximum heart rate, with a correlation coefficient of approximately -0.41.

Overall, this project demonstrates how R can be used to analyze real-world health data and convert it into meaningful statistical insights and visualization.

17. References

Heart Disease Dataset (UCI Machine Learning Repository / Kaggle)

R Programming Documentation
R Markdown Documentation